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A SIMULATION OF SMALL BREAK LOSS OF COOLANT ACCIDENT (SB-LOCA) IN AP1000 NUCLEAR POWER PLANT USING RELAP5-MV

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ABSTRACT

To ensure effective operation of nuclear power plants, it is very important to evaluate different accident scenarios in actual plant conditions with different codes. In the field of nuclear safety, Loss of Coolant Accident (LOCA) is one of the main accidents. RELAP-MV Visualized Modularization software technology is recognized as one of the best estimated transient simulation programs of light water reactors, and also has the options for improved modeling methods, advanced programming, computational simulation techniques and integrated graphics displays. In this study, transient analysis of the primary system variation of thermo-hydraulics parameters in primary loop under SB-LOCA accident in AP1000 nuclear power plant (NPP) is carried out by Relap5-MV thermo-hydraulics code. While focusing on LOCA analysis in this study, effort was also made to test the effectiveness of the RELAP5-MV software already developed. The accuracy and reliability of RELAP5-MV have been successfully confirmed by simulating LOCA. The calculation was performed up to a transient time of 4,500.0s. RELAP5-MV is able to simulate a nuclear power system accurately and reliably using this modular modeling method. The results obtained from RELAP5 and RELAP5-MV are in agreement as they are based on the same models though in comparison with RELAP5, RELAP5-MV makes simulation of nuclear power systems easier and convenient for users most especially for the beginners.

Keywords: Nuclear Safety, Relap5, Loss of Coolant Accident and AP1000

1. INTRODUCTION

The AP1000 is an advanced nuclear power plant. One of the innovative designs of the AP1000 from conventional Westinghouse pressurized water reactors (PWRs) is that it include passive safety features to prevent or minimize core un-covery during small break loss of coolant accidents (SBLOCAs) [1]. The core of AP1000 has 14 feet active fuel length and is rated for a thermal power of 3400MWt. There are two loops in the Reactor cooling system (RCS). Each loop comprises of one hot leg and two cold legs. The conventional pressurized water reactors rely on active equipment such as pumps, fans, diesels, chillers, or other rotating machinery [2]. The passive safety systems depend on natural phenomenon such as gravity, compressed gas and natural circulation. Moreover, under accident condition, evaporation is used to provide for long time reactor cooling while PRHRS could remove the residual heat of the core [3]. In-containment Refueling Water Storage Tank (IRWST), Core Makeup Tank (CMT) and the Accumulators system are the three major sources of water available for cooling the core and for residual heat remover to avoid core uncover [4]. The main features of the passive safety system are the passive containment cooling system (PCS) and passive core cooling system (PXS). The subdivisions of the PXS are the passive